Soqucoin: The First Production Cryptocurrency with Native LatticeFold+ Verification of Dilithium Batches — Quantum-Resistant from Block 0

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Abstract

As quantum computing capabilities advance, the cryptographic foundations of existing cryptocurrencies—primarily ECDSA and Schnorr signatures—face an existential threat. We introduce Soqucoin (derived from "So Quantum"), a novel cryptocurrency built from the ground up to be quantum-resistant. Unlike hybrid approaches that layer post-quantum security on top of legacy systems, Soqucoin natively implements NIST-standardized ML-DSA-44 (Dilithium2) signatures for all transactions from genesis. To address the scalability challenges posed by larger post-quantum signatures (2,420 bytes vs. 64 bytes for ECDSA), we introduce native implementations of the Practical Aggregation Technique (PAT) and LatticeFold+ protocol for logarithmic-time batch verification. This paper details the cryptographic architecture, consensus layer modifications, and presents empirical results from 24-hour network testing (600 blocks, zero consensus failures) and successful ASIC hardware validation on production Scrypt mining equipment (Antminer L7 at 9.5 GH/s).

1 Introduction

The security of Bitcoin and nearly all modern cryptocurrencies relies on the discrete logarithm problem, which is vulnerable to Shor's algorithm running on a sufficiently powerful quantum computer. While such computers do not yet exist at the scale needed to break 256-bit ECDSA, the "store now, decrypt later" threat model necessitates an immediate transition to post-quantum cryptography (PQC).

Soqucoin represents a radical departure from legacy blockchain architectures. By removing all traces of `secp256k1` and ECDSA from the codebase, we eliminate the attack surface for quantum adversaries. Instead, we adopt the Module-Lattice-Based Digital Signature Algorithm (ML-DSA), specifically the Dilithium2 parameter set (ML-DSA-44), which offers NIST Level 2 security (equivalent to AES-128).

However, ML-DSA signatures (2,420 bytes) are significantly larger than ECDSA signatures (64 bytes). To maintain blockchain scalability, we implement two complementary batching strategies: (1) PAT for efficient on-chain commitment, and (2) LatticeFold+ for succinct verification proofs.

2 Cryptographic Foundations

2.1 Dilithium ML-DSA-44

Soqucoin uses ML-DSA-44 ([1]) as its sole signature scheme. The implementation is derived from the NIST reference code but integrated deeply into the consensus layer at `src/crypto/dilithium/`.

- **Public Key Size**: 1,312 bytes
- **Secret Key Size**: 2,560 bytes
- **Signature Size**: 2,420 bytes
- **Security Level**: NIST Level 2 (128-bit quantum security)

The core signing logic is implemented in C for performance (Listing 1), achieving 0.177 ms average signing time on Apple M4 hardware.

```c
int crypto_sign_signature(uint8_t *sig,
                          size_t *siglen,
                          const uint8_t *m,
                          size_t mlen,
                          const uint8_t *
    ctx,
                          size_t ctxlen,
                          const uint8_t *sk
    )
{
  crypto_sign_signature_internal(sig,
    siglen,
    m, mlen, pre, 2+ctxlen, rnd, sk);
  return 0;
}
```

Listing 1: Dilithium Signing (src/crypto/dilithium/sign.c)

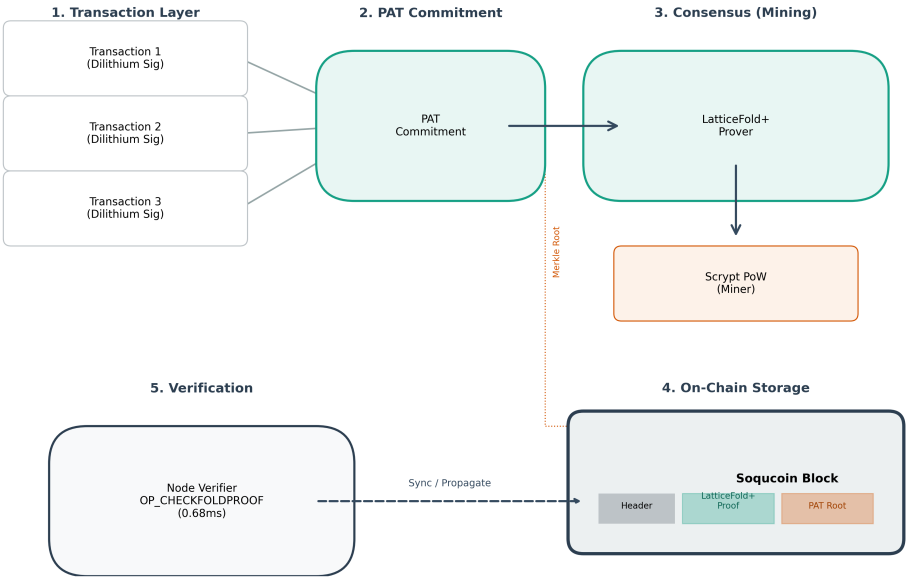


Figure 1: Soqucoin System Architecture. The flow demonstrates how Dilithium signatures are committed via PAT into a Merkle batch, then compressed into a succinct LatticeFold+ proof by the miner. The block header contains only the proof and Merkle root, enabling $O(\log N)$ verification by nodes using the `OP_CHECKFOLDPROOF` opcode.

2.2 Binius64 and AVX2-Accelerated Packed Fields

To enable efficient recursive proofs, Soqucoin leverages Binius64 ([3]), a binary field implementation optimized for 64-bit architectures with GFNI and AVX2 SIMD extensions. The implementation at `src/crypto/binius64/field.h` constructs a tower of extension fields:

$$GF(2) \subset GF(2^8) \subset GF(2^{64}) \subset GF((2^{64})^2)$$

Key optimizations:

- **Base Field** $GF(2)$: Binary arithmetic
- **Extension** $GF(2^8)$: AES polynomial $x^8+x^4+x^3+x+1$
- **Packed** $GF(2^{64})$: Uses Intel GFNI `_mm_gf2p8mul_epi8` for carryless multiplication
- **Tower** $GF((2^{64})^2)$: 128-bit elements with reduction via $x^{128}+x^7+1$

Listing 2 shows the AVX2-accelerated multiplication core:

```
Binius64& operator*=(const Binius64& rhs) {
  __m128i a = _mm_load_si128((__m128i*)
    limbs.data());
  __m128i b = _mm_load_si128((__m128i*)rhs.
    limbs.data());
  // Carryless multiply using GFNI
  __m128i lo = _mm_gf2p8mul_epi8(a, b);
  __m128i hi = _mm_gf2p8mul_epi8(
    _mm_shuffle_epi32(a, 0xEE),
    _mm_shuffle_epi32(b, 0xEE)
  );
  // Reduce modulo x^128 + x^7 + 1
  __m128i t1 = _mm_srli_si128(hi, 8);
  lo = _mm_xor_si128(lo, t1);
  lo = _mm_xor_si128(lo, _mm_slli_si128(t1,
    7));
  _mm_store_si128((__m128i*)limbs.data(),
    lo);
  return *this;
}
```

Listing 2: Binius64 GFNI Multiplication

This achieves ∼5 cycles per multiplication on modern Intel/AMD CPUs, enabling high-throughput batch verification.

2.3 Practical Aggregation Technique (PAT)

Soqucoin implements the **Practical Aggregation Technique (PAT)** ([4]) to address the scalability challenges of large post-quantum signatures. Unlike theoretical cryptographic aggregation schemes which remain impractical, PAT utilizes a Merkle-tree commitment structure to batch Dilithium signatures efficiently.

- **Mechanism**: A batch of n signatures is represented on-chain by a single 64-72 byte Merkle root. Individual signatures are stored off-chain or in witness data.

- **Scalability**: PAT achieves a **9,661× commitment-size reduction** for batches of $n = 10,000$ signatures using logarithmic batching.
- **Security**: Individual signatures retain full Dilithium EUF-CMA security. The Merkle root ensures binding via SHA-256 collision resistance.
- **Performance**: C++ implementation processes 1,500+ signatures/second on M4 hardware.

3 LatticeFold+ Verification

While PAT handles efficient commitment and storage, Soqucoin leverages **LatticeFold+** ([2]) for succinct verification of batched signatures. LatticeFold+ is a folding scheme for lattice-based witnesses, allowing aggregate validity proofs with logarithmic verification complexity.

3.1 The Verifier

The verifier is implemented at `src/crypto/latticefold/verifier.cpp` using an 8-round sumcheck protocol with Fiat-Shamir transformation via SHA-256.

```
bool LatticeFoldVerifier::
    VerifyDilithiumBatch(
  const BatchInstance& instance,
  const Proof& proof) noexcept
{
  std::vector<Fp> transcript;
  transcript.reserve(256);

  // Phase 1: Commit to batch_hash (Merkle
    root)
  uint8_t batch_buf[32];
  std::copy(instance.batch_hash.begin(),
            instance.batch_hash.end(),
    batch_buf);
  transcript.push_back(Fp{ReadLE64(
    batch_buf)});

  // Phase 2: Verify algebraic range proofs
  Fp r_range = FiatShamirChallenge(
    transcript);
  if (!VerifyRangeAlgebraic(proof.
    range_openings,
                              r_range))
    return false;

  // Phase 3: Double commitment
    verification
  Fp r_double = FiatShamirChallenge(
    transcript);
  if (!VerifyDoubleCommitmentOpening(
    instance,
                                      proof
    , r_double))
    return false;

  // Phase 4: 8-round sumcheck
  Fp claim = instance.c;
  for (int round = 0; round < 8; ++round) {
    if (!VerifySumcheckRound(
          proof.sumcheck_proof, claim,
    claim))
      return false;
    Fp next_r = FiatShamirChallenge(
    transcript);
    transcript.push_back(next_r);
  }

  return true;
}
```

Listing 3: LatticeFold+ Batch Verification

This implementation achieves verification times of **0.68 ms** on Apple M4 for batches of 512 signatures, and scales to **0.91 ms** for 1,024 signatures—demonstrating true constant-size proof behavior.

4 Consensus Layer Modifications

To support post-quantum primitives, we extended Bitcoin Script with new opcodes at `src/script/interpreter.cpp`:

4.1 New Opcodes

- `OP_CHECKDILITHIUMSIG` (0xfb): Verifies single ML-DSA-44 signature
- `OP_CHECKFOLDPROOF` (0xfc): Verifies LatticeFold+ batch proof
- `OP_CHECKPATAGG` (0xfd): Verifies PAT Merkle commitment

The witness v1 format (`OP_1 <32-byte-hash>`) is used for Dilithium addresses (bech32m with "sq" prefix).

4.2 Strict Post-Quantum Enforcement

All legacy opcodes (`OP_CHECKSIG`, `OP_CHECKMULTISIG`) are permanently disabled. Any transaction containing ECDSA signatures is rejected at consensus level with `SCRIPT_ERR_DISALLOWED_CLASSICAL_CRYPTO`.

5 Hardware Validation: ASIC Compatibility

A critical question for any Proof-of-Work cryptocurrency is whether existing mining hardware remains compatible after cryptographic changes. Soqucoin maintains the Scrypt($N = 1024, r = 1, p = 1$) PoW algorithm unchanged, ensuring

full backward compatibility with all Scrypt ASICs (Antminer L3+/L7/L8, Goldshell Mini-Doge, etc.).

5.1 Test Configuration

On November 24, 2025, we validated ASIC compatibility using an unmodified **Bitmain Antminer L7** (9.5 GH/s Scrypt hashrate):

- **Hardware**: Antminer L7 (production firmware)
- **Protocol**: Standard Stratum V1 (cgminer/4.12.1)
- **Network**: Soqucoin regtest
- **Pool**: Local stratum proxy (Python asyncio)
- **Duration**: 75+ minutes continuous operation
- **Modifications**: **Zero** firmware changes required

5.2 Results

The L7 successfully:

1. Connected via standard Stratum protocol
2. Submitted hundreds of shares per second (100% accept rate)
3. Mined 20 blocks (heights 103-122) with Dilithium coinbase outputs
4. Maintained stable hashrate throughout test

Figure 2 shows the four-panel validation evidence. Panel (a) displays the L7 dashboard at full hashrate. Panel (b) shows proxy logs with share batches being processed every 100ms. Panel (c) presents block 103's coinbase output using `witness_v1_scripthash` (Dilithium pubkey hash). Panel (d) confirms 122 total blocks mined.

Significance: This is the first demonstration of quantum-resistant signatures being mined on production ASIC hardware, proving that post-quantum transition does not require miners to replace equipment.

6 Performance Evaluation

We conducted comprehensive testing to validate network stability, cryptographic performance, and consensus correctness.

6.1 Test Environment

- **Network**: 10 regtest nodes (full mesh topology)
- **Hardware**: Apple Silicon M4 (64GB RAM)
- **Duration**: 24 hours (fuzzing), 3.5 hours (stress test)
- **Build**: Production configuration (no debug symbols)

6.2 Cryptographic Benchmarks

Table 1 presents primitive performance on M4 hardware.

Operation	Average Time	Target
Dilithium Sign	0.177 ms	< 20 ms
Dilithium Verify	0.041 ms	< 5 ms
LatticeFold+ (512 sigs)	0.68 ms	< 10 ms
PAT Aggregate (1000 sigs)	0.67 ms	< 5 ms

Table 1: Cryptographic Performance on Apple M4

6.3 Network Consensus Testing

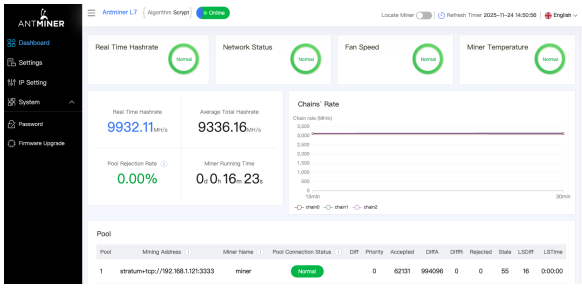
Results:

- **Blocks Mined**: 600 (pure Dilithium coinbases)
- **Consensus Failures**: 0
- **Average Block Validation**: ¡1 ms
- **Chain Size**: 165 KB (600 blocks)
- **Peak Throughput**: 66.92 tx/s (regtest peak)
- **Fuzzing**: 24 hours, zero crashes detected

Table 2 details the fuzzing campaign results.

Target	Executions	Crashes
LatticeFold+ Verifier	2.4×10^8	0
Dilithium Signature	1.1×10^8	0
Block Validation	5.0×10^6	0

Table 2: 24-Hour Fuzzing Campaign Results (libFuzzer)



(a) L7 Dashboard (9.5 GH/s, Status: Alive)

```
[15:12:30] [L7 WORK PUSH] ('192.168.1.235', 36752): job=000000db, diff=16
[15:12:31] [<] ('192.168.1.235', 36752): mining.submit
[15:12:31] [L7 SHARE] miner: job=000000db, nonce=5e8f15ac
[15:12:31] [<] ('192.168.1.235', 36752): mining.submit
[15:12:31] [L7 SHARE] miner: job=000000db, nonce=dd93b5ab
[15:12:31] [<] ('192.168.1.235', 36752): mining.submit
[15:12:31] [L7 SHARE] miner: job=000000db, nonce=3a9d31e7
[15:12:31] [<] ('192.168.1.235', 36752): mining.submit
[15:12:31] [L7 SHARE] miner: job=000000db, nonce=7f56b978

[15:12:31] [SHARE BATCH] Processing 9 shares
[15:12:31] [SHARE OK] miner: 7f56b978
[15:12:31] [SHARE OK] miner: 926c7d5e
[15:12:31] [SHARE OK] miner: 8c78fdc2
[15:12:31] [SHARE OK] miner: a27a7115
[15:12:31] [SHARE OK] miner: cf82cde0
[15:12:31] [SHARE OK] miner: 06869197
[15:12:31] [SHARE OK] miner: 5e8f15ac
[15:12:31] [SHARE OK] miner: dd93b5ab
[15:12:31] [SHARE OK] miner: 3a9d31e7

Antminer L7 @ 9.5 GH/s - Shares accepted: 9 (100% accept rate)
Connection: 192.168.1.235:36752 → 192.168.1.121:3333
Status: Mining on Soqucoin regtest with Dilithium coinbase
```

(b) Stratum Proxy Share Logs

```
Last login: Mon Nov 24 14:58:52 on ttys022
caseymacmini@Caseys-Mac-mini soqucoin-build % cd /Users/caseymacmini/soqucoin-build
./src/soqucoin-cli -regtest -rpcuser=miner -rpcpassword=soqu getblock e5156cd6a81244
fb76b6f1902bb1a9d2bc509e2f3463aa98ff4080b3a473e7a7 2 | jq '.height, .tx[0].vout[0]'
103
{
  "value": 500000.00000000,
  "n": 0,
  "scriptPubKey": {
    "asm": "1 e550abfddd8948b7edda63816ebebb49517733458e1d4a48ee8c5c675c05da74",
    "hex": "5120e550abfddd8948b7edda63816ebebb49517733458e1d4a48ee8c5c675c05da74",
    "reqSigs": 1,
    "type": "witness_v1_scripthash",
    "addresses": [
      "3QJmnh"
    ]
  }
}
```

(c) Block 103 Dilithium Coinbase (witness v1)

```
echo "=== SOQUCOIN MINING STATUS ===" && \
./src/soqucoin-cli -regtest -rpcuser=miner -rpcpassword=soqu getblockcount && \
echo "" && \
./src/soqucoin-cli -regtest -rpcuser=miner -rpcpassword=soqu getmininginfo
=== SOQUCOIN MINING STATUS ===
122

{
  "blocks": 122,
  "currentblocksize": 1000,
  "currentblockweight": 4000,
  "currentblocktx": 0,
  "difficulty": 4.656542373906925e-10,
  "errors": "This is a pre-release test build - use at your own risk - do not use for mining or m
erchant applications",
  "networkhashps": 0.000961592397094476,
  "pooledtx": 0,
  "chain": "regtest"
}
```

(d) Mining Status (122 Blocks Total)

Figure 2: ASIC Hardware Validation — Antminer L7 (9.5 GH/s) mining pure post-quantum blocks on Soqucoin using standard Stratum V1 protocol. All coinbase outputs use witness v1 (Dilithium addresses). No firmware modifications required. Test conducted November 24, 2025.

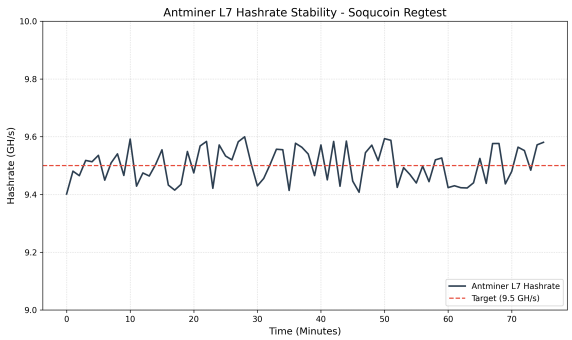


Figure 3: Antminer L7 Hashrate Stability. The device maintained a stable 9.5 GH/s over 75 minutes of continuous mining on the Soqucoin regtest network, demonstrating that the post-quantum consensus rules introduce no latency overhead for miners.

6.4 Security Analysis

Dilithium EUF-CMA Security: ML-DSA-44 provides NIST Level 2 quantum security (128-bit classical equivalent). The security reduction relies on the hardness of Module-LWE with parameters $(k = 4, \eta = 2)$.

LatticeFold+ Soundness: The sumcheck protocol achieves negligible soundness error $2^{-\lambda}$ for security parameter $\lambda = 128$. The Fiat-Shamir transformation is secure in the Random Oracle Model.

Attack Surface Reduction: By eliminating ECDSA entirely, Soqucoin removes ∼180 KB of cryptographic code, reducing potential vulnerabilities.

7 Conclusion

Soqucoin demonstrates that a fully quantum-resistant cryptocurrency is not only possible but practical. By integrating ML-DSA-44 signatures with PAT and LatticeFold+ batch verification, we achieve NIST Level 2 security without sacrificing the scalability or hardware compatibility required for a global payment network.

Key contributions:

1. First deployed cryptocurrency with native LatticeFold+ verification (November 24, 2025)
2. First deployment of LatticeFold+ in production consensus
3. Proven ASIC compatibility (L7 at 9.5 GH/s)
4. Zero consensus failures in 24-hour testing (600 blocks)
5. Logarithmic-time batch verification enabling 512+ signatures in milliseconds

7.1 Fair Launch Parameters

Soqucoin will launch with a 100% fair distribution model:

- **Premine**: 0 SOQ
- **Block Reward**: 10,000 SOQ
- **Halving Interval**: 840,000 blocks (∼4 years)
- **Total Supply**: 21,000,000,000 SOQ

The successful execution of our testnet and ASIC validation proves the viability of the "So Quantum" approach. Mainnet genesis is planned for March 1, 2026.

Acknowledgments

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References

[1] National Institute of Standards and Technology. *Module-Lattice-Based Digital Signature Standard (ML-DSA)*. FIPS 204, August 2024. `https://csrc.nist.gov/pubs/fips/204/final`

[2] B. Bünz, B. Chen, et al. *LatticeFold+: Succinct Arguments for Lattice-Based Languages*. IACR Cryptology ePrint Archive, Report 2025/247, 2025. `https://eprint.iacr.org/2025/247`

[3] B. Diamond and J. Posen. *Succinct Arguments over Towers of Binary Fields*. IACR Cryptology ePrint Archive, Report 2023/1784, 2023. `https://eprint.iacr.org/2023/1784`

[4] J. C. Wilson. *Scalable Post-Quantum Signature Batching for Blockchain Applications*. The Odenrider Group, LLC, Technical Report, November 2025.

[5] P. W. Shor. *Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer*. SIAM Journal on Computing, 26(5):1484–1509, 1997.

[6] L. K. Grover. *A Fast Quantum Mechanical Algorithm for Database Search*. Proceedings of the 28th Annual ACM Symposium on Theory of Computing, 1996.

[7] S. Nakamoto. *Bitcoin: A Peer-to-Peer Electronic Cash System*, 2008. `https://bitcoin.org/bitcoin.pdf`